

DNNs and human observers make different mistakes when classifying natural images.

Does this happen because humans have too much time to think about the images?

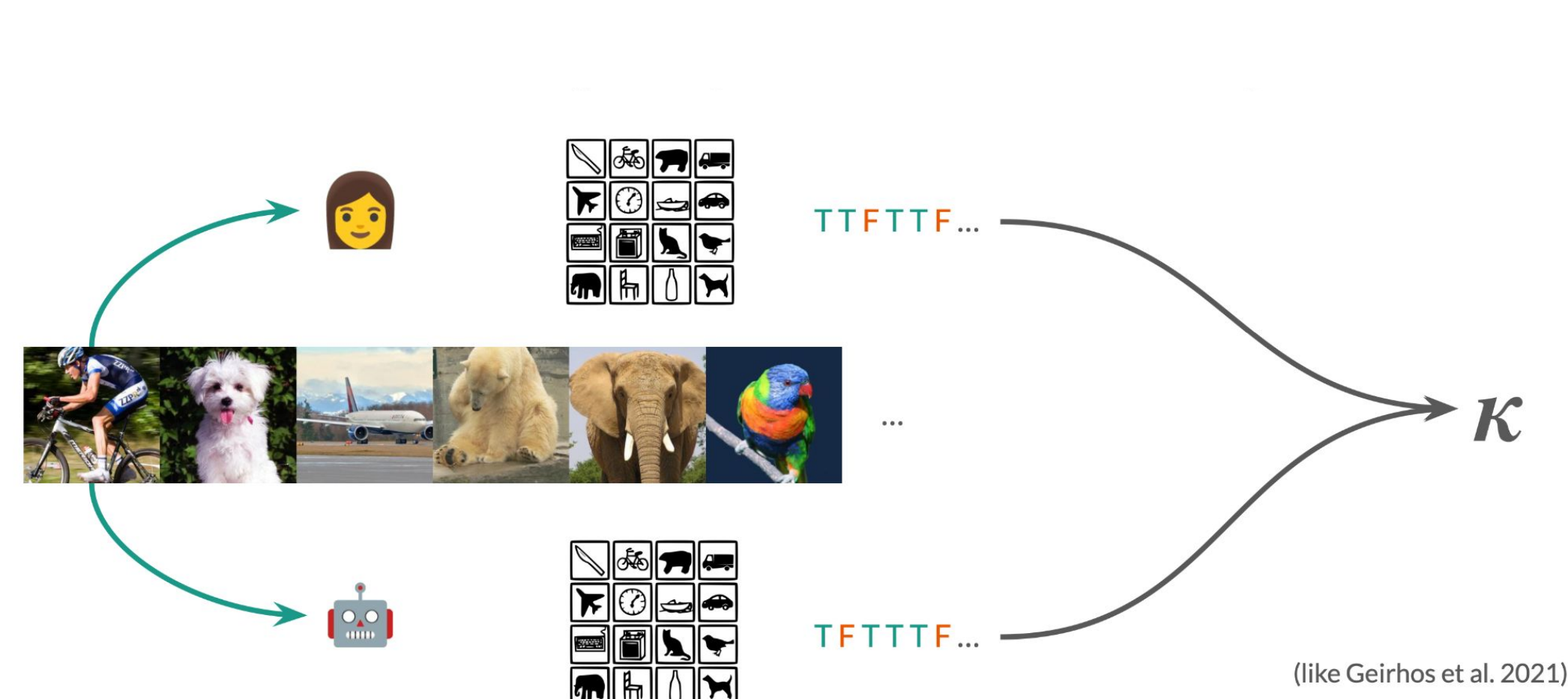
No! Even at very short presentation times, error consistency between humans and machines is low.

Behavioral Differences in Rapid Object Recognition between Humans and Machines

Thomas Klein, Wieland Brendel and Felix A. Wichmann

Background

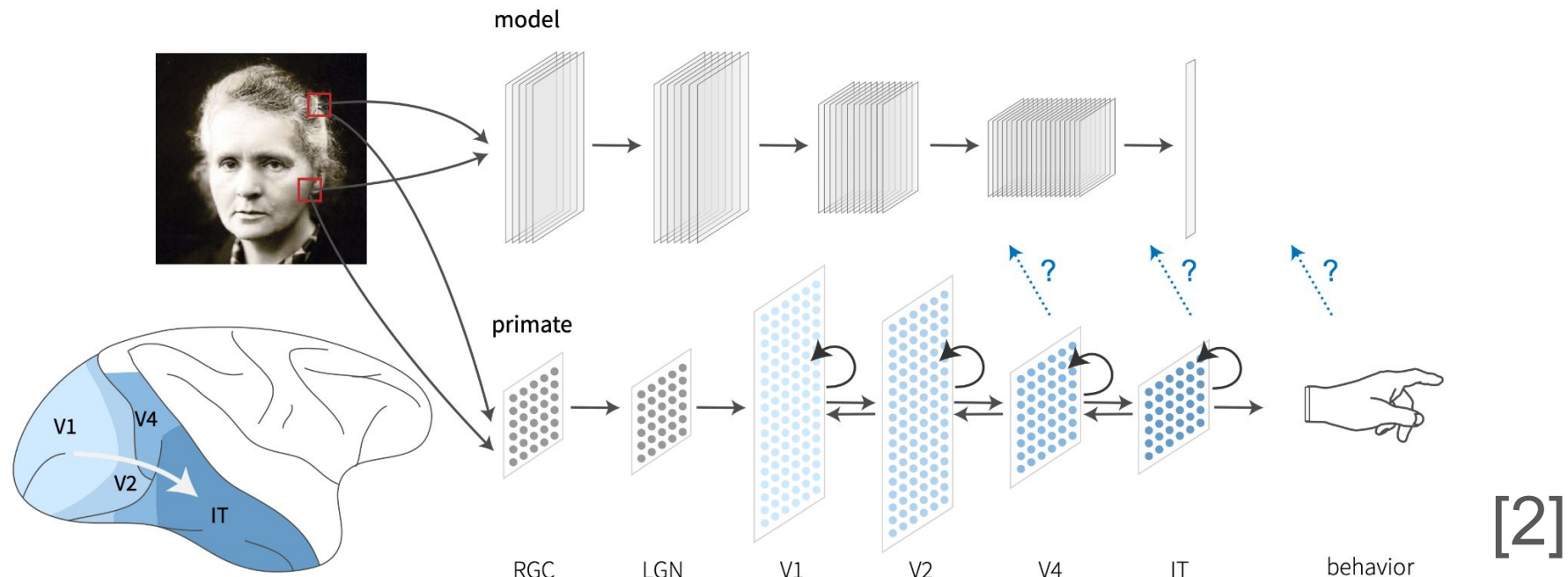
Earlier work [1] finds that DNNs make different mistakes than humans, as quantified by *Error Consistency*:



	T	F
T		
F		

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

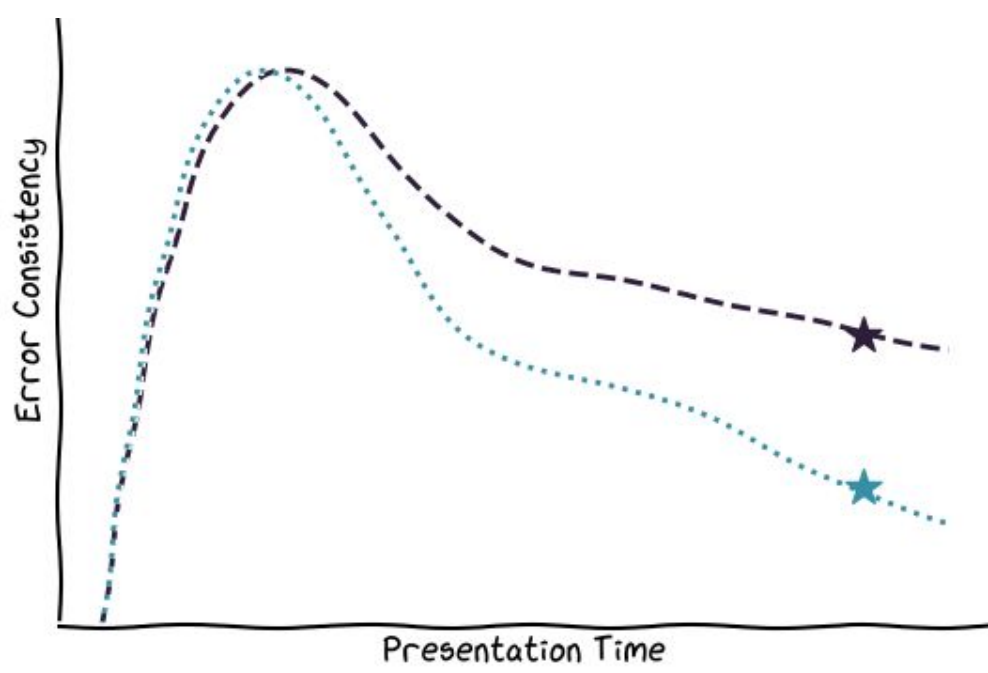
But at the same time, DNNs are considered to be the best model of the human visual system [2]:



So maybe, DNNs are only good models of the ventral stream for the forward sweep, but then higher-level (recurrent) processing leads to behavioral differences.

Implication:

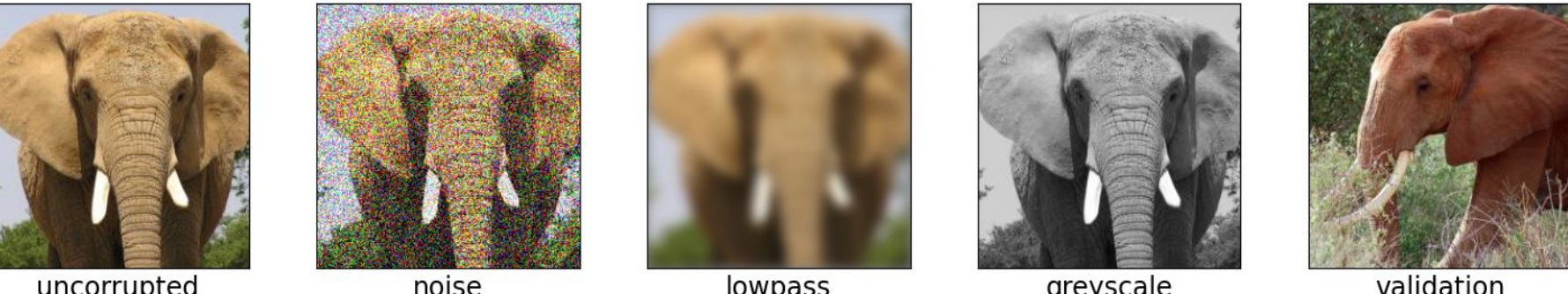
Higher EC at shorter presentation times!



Experiment

We measure EC between humans and machines, with presentation times from 8ms to 266ms.

We apply corruptions to make images harder:



We manually clean the images we use for our experiments, to avoid problematic stimuli:



Experimental Setup

We show images on a TouchPixx-display, to collect responses as quickly as possible.

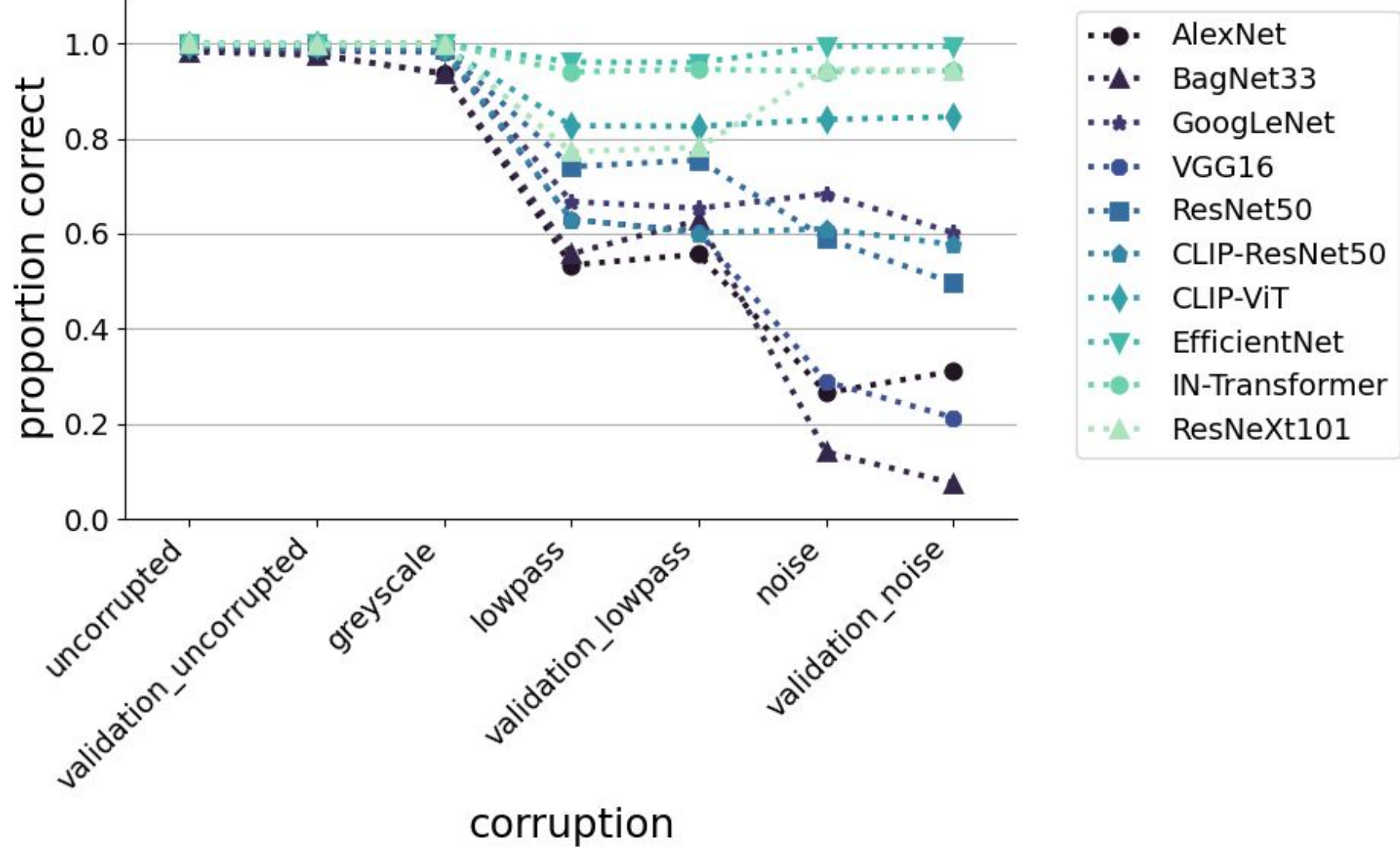


We test nine subjects (five female) with normal vision, each for about four hours across two sessions on different days.

Experiments are conducted in a darkened lab.

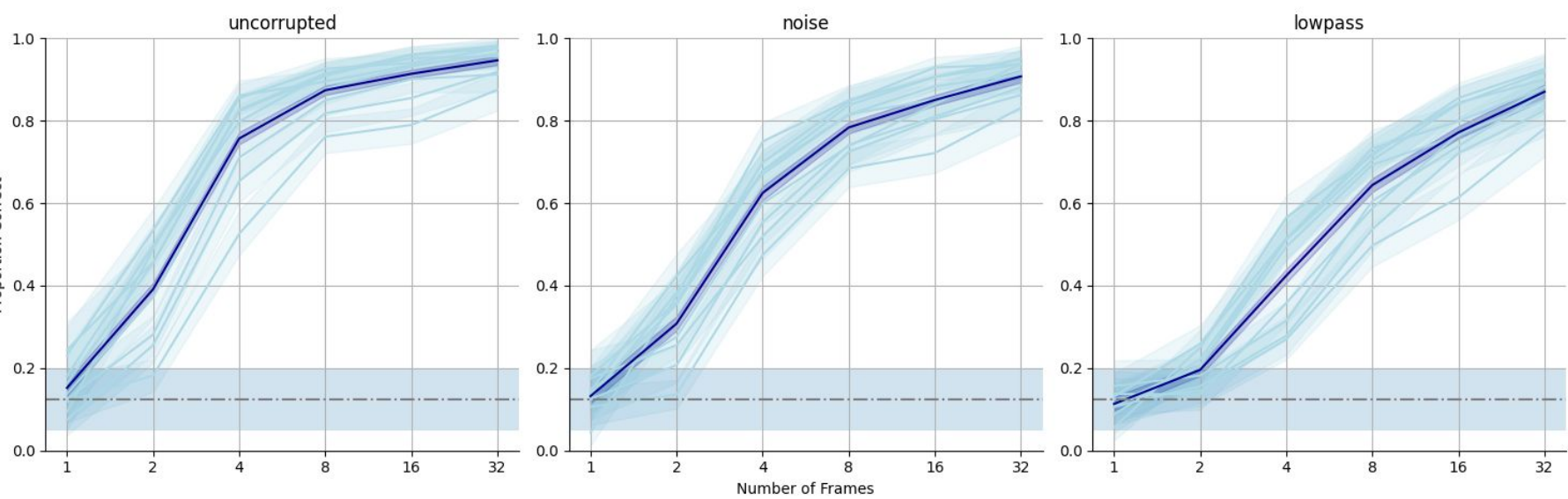
Deep Neural Networks

We compare our human subjects to a suite of 10 DNNs (5 classic CNNs, 4 modern models, BagNet as sanity check).

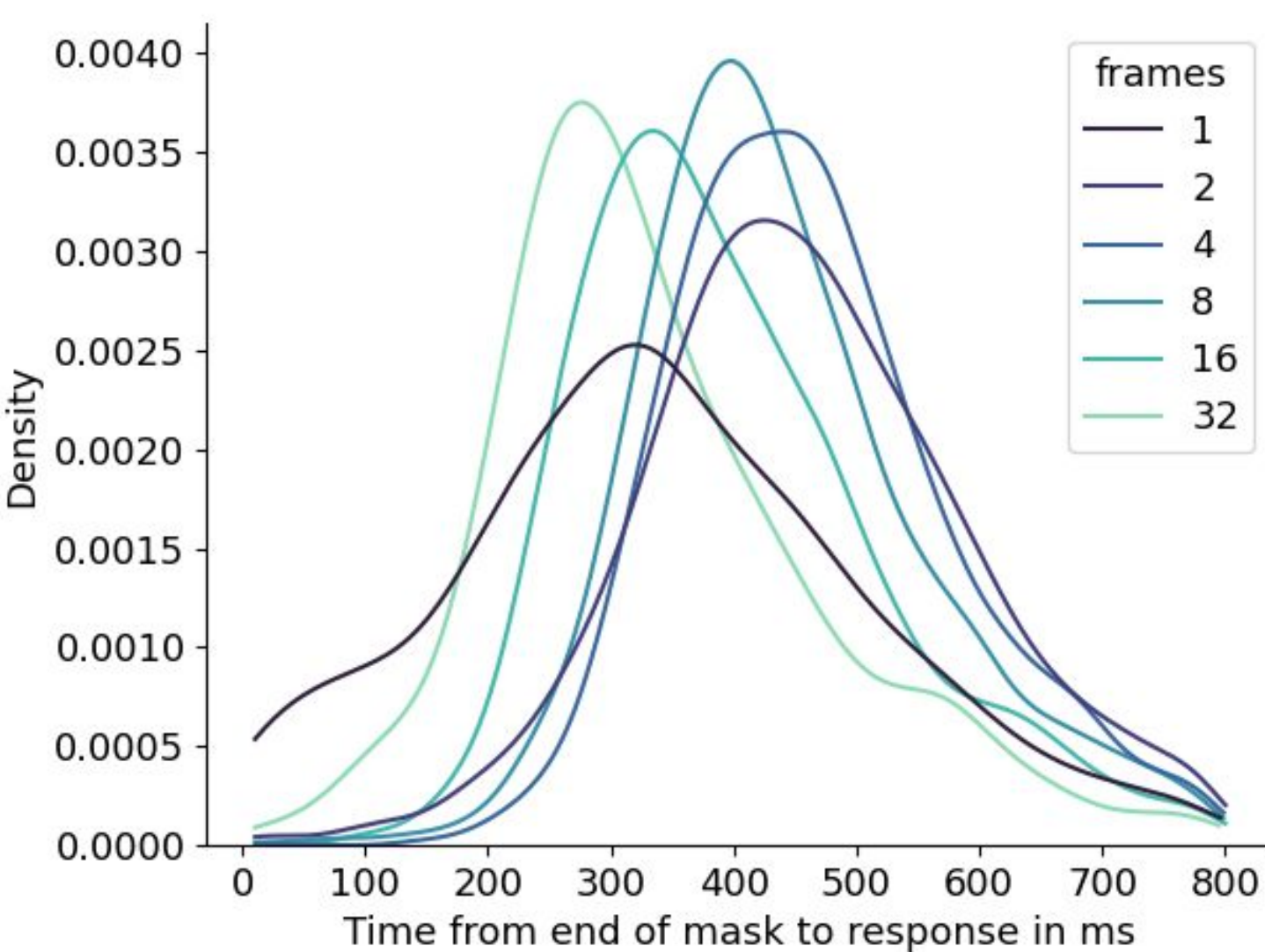


Human Performance

Humans benefit from more presentation time, but can solve the task very early:

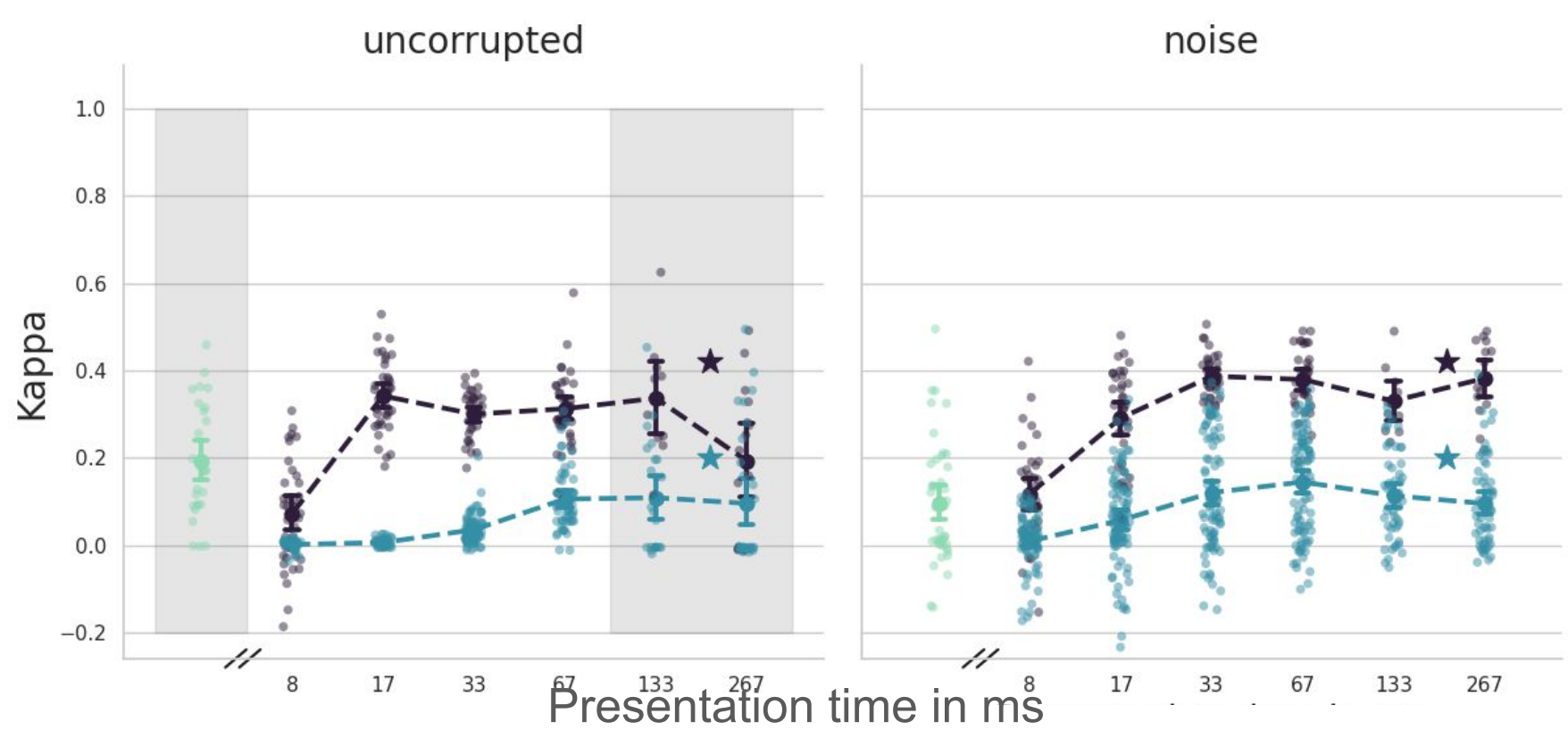
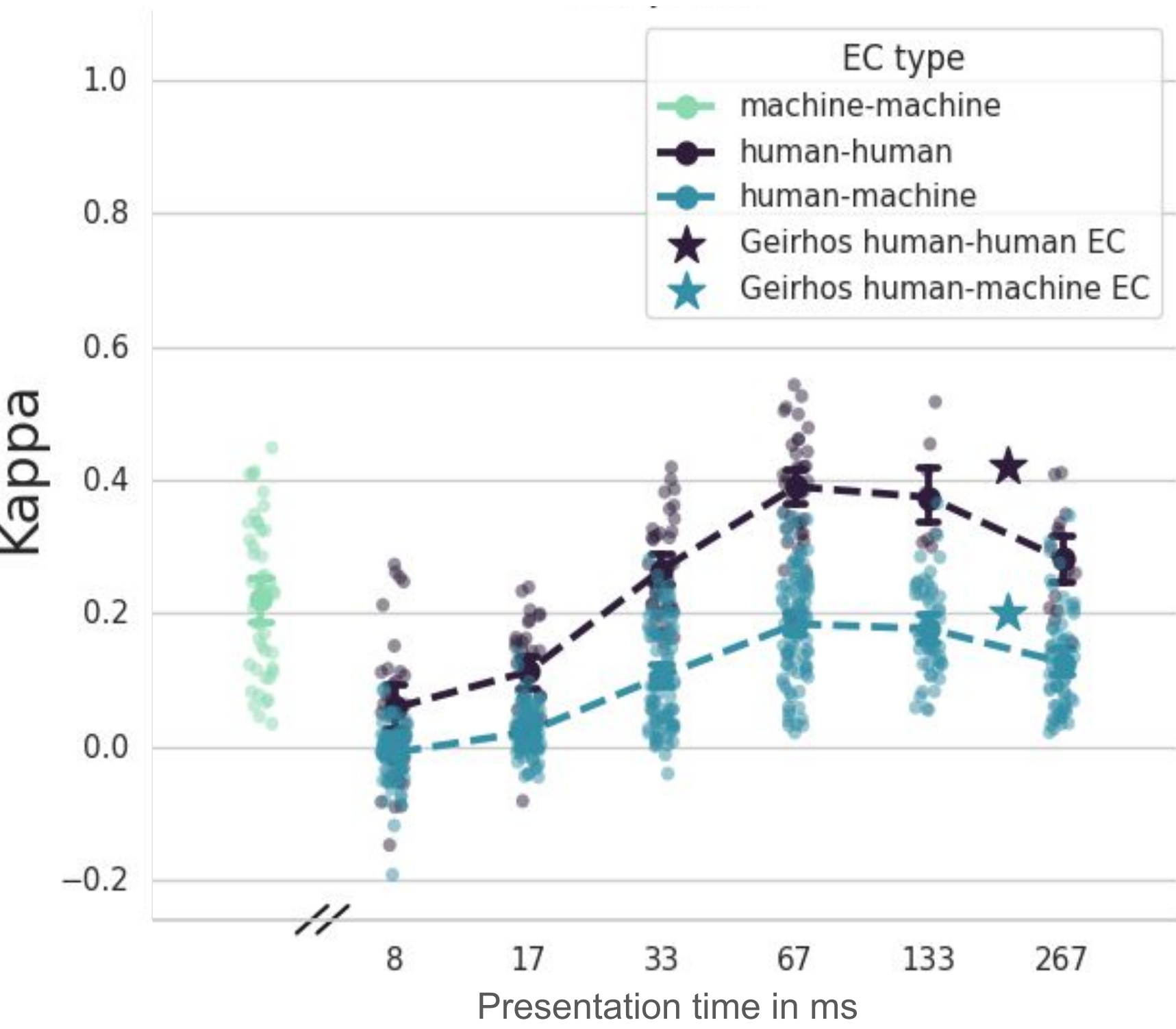


Humans are also able to solve this task *very quickly*, with a median RT of 668 ms (about 195 ms faster than [1], so very little time for recurrent processing).



Error Consistency

EC never exceeds values known from earlier work:



Methodological Insights

1. It matters whether you use training set or validation set images to evaluate EC.

AlexNet*	0.40	0.31	0.36	0.17	0.00	0.09	0.17	0.13	0.00
BagNet33*	0.38		0.36	0.32	0.08	0.00	0.12	0.09	0.05
GoogLeNet*	0.32	0.43		0.46	0.26	0.00	0.28	0.09	0.20
VGG16*	0.28	0.41	0.56		0.18	0.00	0.31	0.12	0.09
ResNet50*	0.13	0.21	0.31	0.50		0.00	0.36	-0.00	-0.00
IN-Transformer*	-0.00	0.08	0.12	0.20	0.50		0.00	0.00	0.00
ResNeXt101	0.06	0.20	0.30	0.46	0.28	0.40		-0.00	0.20
CLIP-ResNet50	0.29	0.25	0.32	0.34	0.35	0.13	0.33		0.17
CLIP-ViT	0.06	0.14	0.31	0.33	0.67	0.50	0.57	0.35	
EfficientNet	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

2. Error Consistency has a few issues, it is...

- ... undefined for two perfect raters
 - ... zero even for one perfect rater
 - ... agnostic to the number of classes
 - ... blind to different types of mistakes
 - ... not always bounded to [-1,1]
- We are working on a tutorial paper that explains these issues, and will try to fix them!

Acknowledgements

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References

- [1] Geirhos et al. (2021). Partial success in closing the gap between human and machine vision. *Advances in Neural Information Processing Systems*, 34, 23885-23899.
- [2] Schrimpf, M., Kubilius, J., Hong, H., Majaj, N. J., Rajalingham, R., Issa, E. B., ... & DiCarlo, J. J. (2018). Brain-score: Which artificial neural network for object recognition is most brain-like?. *BioRxiv*, 407007.

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